Scaredy Cat

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Scaredy Cat is a volume shaper that makes your sound duck, pump or chop in time with the music, without needing a sidechain. It splits the sound into three bands (low, mid, high) and gives each band its own drawn volume curve that repeats every beat, bar or note you choose. Use it for the kick pump in hardstyle and house, trance gates, UK hardcore chops, tremolo, or gentle glue. AMOUNT 0 % is an exact bypass; AMOUNT 100 % is the full depth you drew.

Quick start

• Pick a preset close to what you want (Hardstyle kick pump, Sidechain house, Trance gate 1/16, UK hardcore chop).
• Press play. The red line in the lanes shows where the cycle is. Set Speed to the note length you want (1/4 = one cycle per beat).
• Draw: drag the round points, double click a lane to add a point, right click a point to delete it. Hold Alt and drag up or down on a lane to bend a segment.
• Raise or lower AMOUNT for the overall depth, and the Low, Mid and High knobs to leave a band alone (0 %) or fully shaped (100 %).
• If the pump is early or late against your kick, move Offset until it lines up.
• Prefer the sound of a hit, not the tempo? Set Trigger to Input hit: every loud hit restarts the envelope.

How the window works

• Three lanes: HIGH on top, MID in the middle, LOW at the bottom. Each lane is the volume of that band over one cycle; the scale on the left is in dB (top is 0 dB, no change).
• The faint grey bars behind the lanes are your input folded over the cycle, so you can see where the kick hits.
• Under the lanes: shape buttons (Pump, Gate, Swell, Ramp up, Step, Wave, Ramp down, V) and More (trance gate, offbeat pump, random steps, tremolo, reverse, flip, copy to all lanes).
• The tool row above the lanes: select and move, draw a ramp up, ramp down, hold or step by dragging across a lane, delete, select all, and move the selection or the whole lane left and right by one grid step.
• Snap, Grid and Swing sit at the right end of the tool row. Hold Ctrl while dragging to move freely.
• Left: the crossover graph (drag the white lines to move the crossovers), Slope, the three band amounts with Solo buttons, Linked or Unlinked envelopes, Linear phase, and the sidechain options.
• Band spectrum: the sound by frequency at each band's range, dashed before the ducking and filled after. Switch to Level to see each band's peak level in dB.
• Right: the big AMOUNT knob, the output level (OUT) and the biggest gain reduction (GR) in dB, output meters and one gain reduction meter per band.
• Bottom bar: Speed (a note length when Sync is on, Hz when off), Trigger, Drive, Limit, Smooth and Offset. The padlock locks the envelopes so a stray click cannot change them.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• Save stores your settings and all three envelopes as your own preset in the Studio 87 presets folder; they appear in the preset menu after the factory presets.
• Resize the window from its bottom right corner; the 100% button returns to the normal size.

Controls

Controls

Power	Choice: Off / On (default On)	Turns the effect on or off. Off is an exact bypass.
Sync	Choice: Hz / Sync (default Sync)	Sync follows the host tempo. Hz runs at a fixed speed.
Trigger	Choice: Repeat / Input hit (default Repeat)	Repeat: the envelope loops with the tempo. Input hit: every loud hit in the input restarts it (no tempo needed).
Slope	Choice: 12, 24, 48 (default 24)	Steepness of the band split in dB per octave.
Multiband	Choice: Off / On (default On)	On: three bands with their own envelopes. Off: one envelope for the whole sound.
Linear phase	Choice: Off / On (default Off)	Splits the bands without phase shift. Adds latency (reported to your host) and a softer low crossover.
Drive type	Choice: Tube, Tape, Soft (default Tube)	Character of the drive: Tube is warm and even, Tape is gentle, Soft is smooth and round.
Limiter	Choice: Off / On (default On)	Lookahead limiter after the ducking.
Offset unit	Choice: MS / SAMPS (default MS)	Offset in milliseconds or in samples.
Envelopes	Choice: Unlinked / Linked (default Unlinked)	Linked: all three bands use the same envelope (edit one, all follow).
Solo	Choice: Off, Low, Mid, High (default Off)	Listen to one band only.
Display	Choice: Input, Sidechain, Both (default Sidechain)	What the dashed spectrum line shows: the input, the sidechain, or both.
Sidechain triggers	Choice: Off / On (default Off)	The sidechain input (if connected) restarts the envelope on every hit.
Grid	Choice: 1/4, 1/8, 1/16, 1/32 (default 1/16)	Grid for editing the envelope.
Snap	Choice: Off / On (default On)	Points snap to the grid while you drag.
Amount	0 to 100 % default 100 %	How deep the envelopes cut. 0 % is an exact bypass (the sound passes untouched), 100 % is the full depth you drew.
Low	0 to 100 % default 100 %	How much of the low band envelope is applied (0 % leaves the low band alone).
Mid	0 to 100 % default 100 %	How much of the mid band envelope is applied.
High	0 to 100 % default 100 %	How much of the high band envelope is applied.
Low crossover	40 to 400 Hz default 120 Hz	Where the low band ends. Drag the line in the crossover graph or the spectrum.
High crossover	800 to 8000 Hz default 2400 Hz	Where the high band starts.

Speed (Hz)	0.05 to 30 Hz default 2 Hz	Speed in cycles per second when Sync is off.
Swing	0 to 100 % default 0 %	Delays the second half of every cycle for a swung feel.
Smooth	0 to 100 % default 4 %	Rounds the envelope edges so sharp steps do not click. Higher is softer.
Drive	0 to 100 % default 0 %	Smooth saturation before the ducking (oversampled). The level stays about the same, the sound gets thicker.
Limit ceiling	-12 to 0 dB default -0.3 dB	Highest level the limiter lets through.
Offset	-100 to 100 default 0	Moves the envelope later (positive) or earlier (negative), in ms or samples. Use it to line the pump up with the kick.
Hit threshold	-60 to 0 dB default -24 dB	How loud a hit must be to restart the envelope in the hit trigger modes.

Recommendations

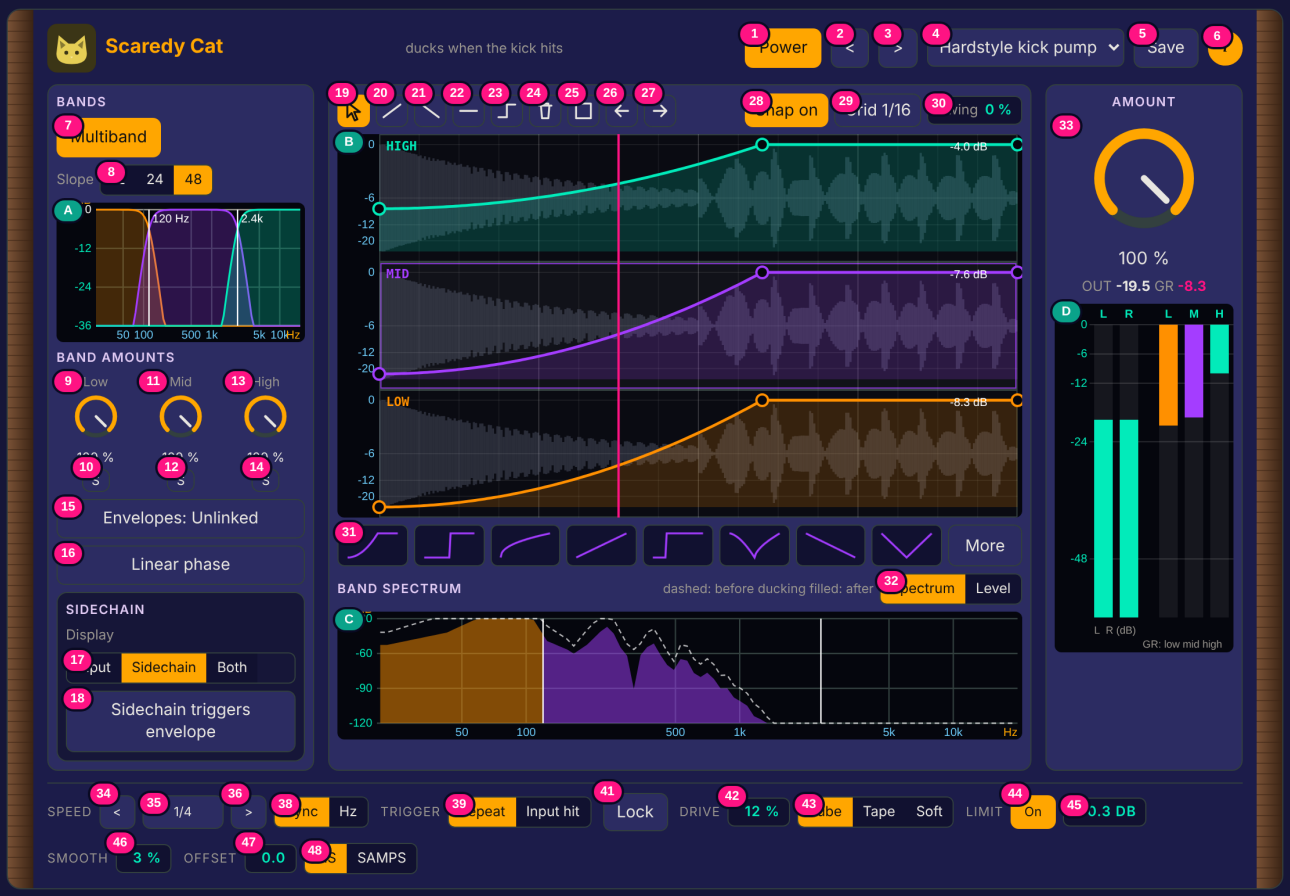
• Kick pump: Speed 1/4, shape Pump, Smooth 3 to 6 %. Keep the High band shallower than the Low band so the hats stay open while the bass ducks.
• Bass only: set Mid and High to 0 % (or use the Bass-only duck preset) so only the low band pumps.
• Trance gate: More, Trance gate x4, Speed 1/4 for sixteenth-note gates; raise Smooth until the clicks disappear.
• No tempo, no kick to follow: Sync off and use Hz, or Trigger Input hit with the threshold just under your kick.
• Smooth steps: sharp steps always click a little; Smooth rounds them. Linear phase is cleaner between bands but adds latency (about 8 to 24 ms at 48 kHz, depending on Slope), which your host compensates.
• Compare with the plugin off (Power) at the same loudness: pumping makes things sound louder or quieter than they are.
• Latency: Scaredy Cat reports its exact latency to your host and updates it when you switch Linear phase or the slope.

Presets

Default, Hardstyle kick pump, Sidechain house, Trance gate 1/16, Eighth-note pump, UK hardcore chop, Slow tremolo, Gentle glue, Half-time swell, Audio hit duck, Bass-only duck

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is taken from the interactive test page of the plugin, which mirrors the plugin window; the compiled plugin may differ by a few pixels of layout.



Signal flow

Input > Drive > band split (low, mid, high; 12, 24 or 48 dB per octave, or linear phase) > one drawn volume envelope per band > Amount blend > lookahead limiter > hidden soft clipper > output. Power off or Amount 0 % is an exact bypass (delayed by the reported latency).

Header bar

1	**Power**	Power. Off is an exact bypass: the sound passes untouched.
2	**Previous preset**	Previous preset.
3	**Next preset**	Next preset.
4	**Preset menu**	Factory presets, then your own (Save).
5	**Save**	Save the current settings and the three envelopes as your own preset (opens the preset box under the plugin).
6	**Info bubbles (i)**	Turn these hover tips on or off.

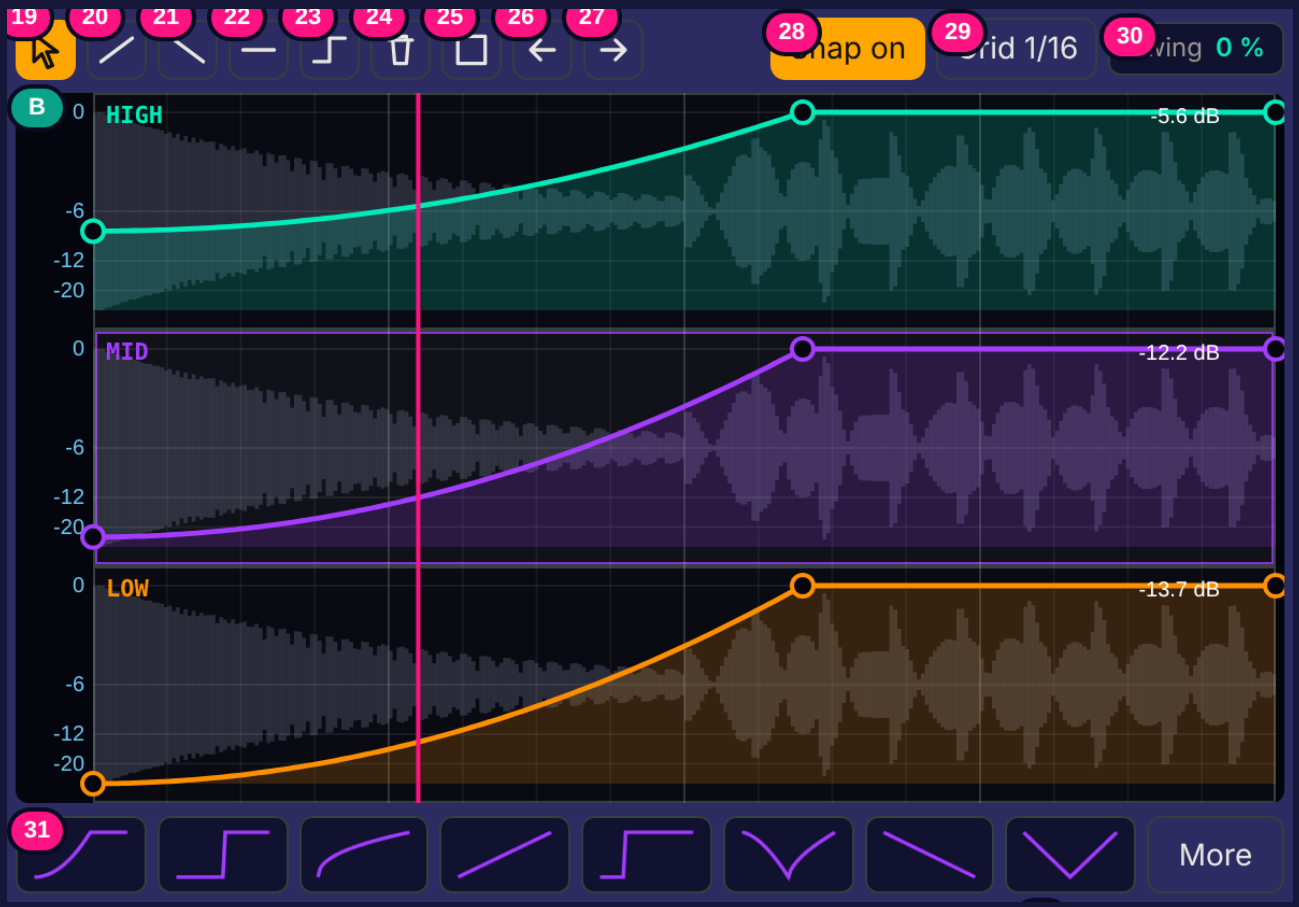
A. Bands and sidechain



Display A: Crossover graph. Horizontal: frequency (log, 20 Hz to 20 kHz). Vertical: level of each band in dB (0 to -36). Orange is the low band, purple the mid band, mint the high band. The two white lines are the crossover points (Low about 120 Hz, High about 2.4 kHz by default): drag them. The Slope strip changes how steep the bands overlap.

7	**Multiband**	Multiband on: three bands with their own envelopes. Off: one envelope for everything (the middle lane).
8	**Slope (12, 24, 48 dB per octave)**	How steep the split between the bands is, in dB per octave.
9	**Low amount**	How much of the low band envelope is applied. 0 % leaves that band alone. (shown: 100 %)
10	**Solo**	Solo: listen to this band only.
11	**Mid amount**	How much of the mid band envelope is applied. 0 % leaves that band alone. (shown: 100 %)
12	**Solo**	Solo: listen to this band only.
13	**High amount**	How much of the high band envelope is applied. 0 % leaves that band alone. (shown: 100 %)
14	**Solo**	Solo: listen to this band only.
15	**Envelopes Linked or Unlinked**	Linked: all three bands use the same envelope, edit one and all follow.
16	**Linear phase**	Linear phase split: no phase shift between the bands. Adds latency (your host compensates) and a softer low crossover.
17	**Sidechain display**	What the dashed spectrum line shows: the input, the sidechain, or both.
18	**Sidechain triggers envelope**	Hits on the sidechain input restart the envelope (pick a sidechain sound in the bar above).

B. Envelope editor



Display B: The three envelope lanes: HIGH on top, MID in the middle, LOW at the bottom. Each lane is the volume of that band over one cycle (left to right). The scale on the left is in dB (top is 0 dB, no change; the lane is drawn linear in gain). Vertical grid lines follow the Grid setting. The grey bars behind the curve are the input folded over one cycle, so you see where the kick hits. The pink line is the playhead; the number on the right of each lane is the gain reduction of that band right now. Drag round points, double click to add one, right click to delete, Alt and drag to bend a segment.

19	**Tool: select and move**	Select and move points.
20	**Tool: ramp up**	Draw a ramp up: drag across the lane.
21	**Tool: ramp down**	Draw a ramp down: drag across the lane.
22	**Tool: hold**	Draw a flat hold: drag across the lane.
23	**Tool: step**	Draw a step: hold the level, then jump.
24	**Tool: delete or reset lane**	Delete the selected points (nothing selected: reset the lane).
25	**Tool: select all**	Select all points in the lane.
26	**Tool: move left**	Move the selection (or the whole lane) one grid step left.
27	**Tool: move right**	Move the selection (or the whole lane) one grid step right.
28	**Snap**	Points snap to the grid while you drag (hold Ctrl to move freely). Click to switch.
29	**Grid**	Grid size for editing. Click to change.
30	**Swing**	Delays the second half of each cycle for a swung feel. Drag up or down.

31	**Shape buttons and More**	Quick shapes for the selected lane. More has trance gates, tremolo, random steps and tools.

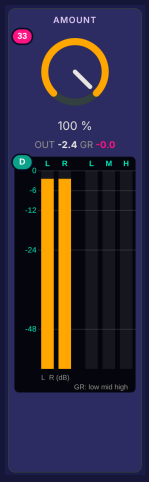
C. Band spectrum



Display C: Band spectrum. Spectrum view: horizontal frequency (log), vertical level in dB; the dashed white line is the sound before ducking and the coloured fill is the sound after, split in the three band colours at the crossover lines (drag them here too). Level view: grey bars are each band's peak level before the ducking and the coloured bars after, in dB.

32	**Spectrum or Level view**	Spectrum: the sound by frequency, dashed before the ducking and filled after. Level: peak level of each band.

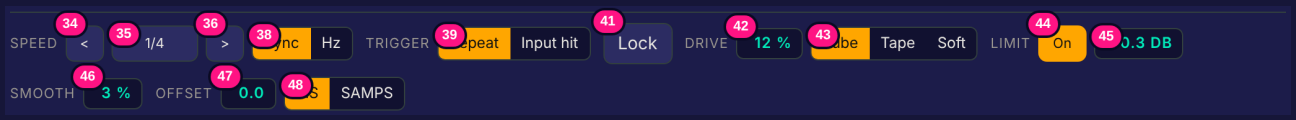
D. Amount and meters



Display D: Meters. L and R: output peak in dB. The three bars on the right (L, M, H): gain reduction of the low, mid and high band, growing downward, 0 to -24 dB. Above them, OUT is the output peak and GR the biggest gain reduction at this moment.

33	**AMOUNT**	AMOUNT: how deep the envelopes cut. 0 % is an exact bypass, 100 % is the full depth you drew. (shown: 100 %)

Bottom bar



34	**Speed slower**	Slower.
35	**Speed (note length)**	How fast the envelope repeats: a note length in Sync mode, or Hz.
36	**Speed faster**	Faster.
37	**Speed in Hz (only when Sync is off)**	Speed in cycles per second when Sync is off. Drag up or down.
38	**Sync or Hz**	Sync follows the host tempo (150 BPM here). Hz runs at a fixed speed.
39	**Trigger**	Repeat: loops with the tempo. Input hit: every loud hit restarts the envelope (no tempo needed).
40	**Hit threshold (only for Input hit)**	How loud a hit must be to restart the envelope. Drag up or down.
41	**Lock**	Lock the envelopes so a stray click cannot change them.
42	**Drive**	Smooth saturation before the ducking (oversampled). The level stays about the same, the sound gets thicker. Drag up or down.
43	**Drive type**	Drive character: Tube is warm, Tape is gentle, Soft is round.
44	**Limiter on**	Lookahead limiter after the ducking.
45	**Limit ceiling**	Highest level the limiter lets through. Drag up or down.
46	**Smooth**	Rounds the envelope edges so steps do not click. Drag up or down.
47	**Offset**	Moves the envelope later (+) or earlier (-) so the pump lines up with your kick.
48	**Offset unit**	Offset in milliseconds or in samples.

Under the hood

• AMOUNT 0 % (or Power off) is an exact bit-exact bypass at the reported latency. 100 % is the full depth you drew: a drawn envelope that goes to the bottom ducks about 64 dB; 50 % ducks about 6 dB. A flat envelope leaves the tone within 0.25 dB.
• Speed: in Sync mode the cycle is a note length of the host tempo (2 bars down to 1/32, triplets and dotted values); in Hz mode it runs free from 0.05 to 30 Hz. Trigger Input hit restarts the envelope on every loud hit (threshold in dB, 50 ms hold-off), no tempo needed. The test page plays at 150 BPM.
• Envelopes are curves of up to 128 points with a bend value per segment. Smooth rounds the edges (about 0.3 ms at 0 % up to about 20 ms at 100 %). Swing delays the second half of each cycle. Offset moves the whole envelope in ms or samples.
• Band split: 12, 24 or 48 dB per octave (Linkwitz-Riley style, bands add back flat). Linear phase uses FIR filters of 513, 1025 or 2049 taps: no phase shift between bands, softer low crossover, more latency.
• Latency is reported to the host and is exact: 64 samples plus the 1.5 ms limiter lookahead (131 samples at 44.1 kHz), and for linear phase plus 256, 512 or 1024 samples (392, 648 and 1160 samples at 48 kHz in total).
• Drive (Tube, Tape, Soft) runs 2x oversampled before the ducking and keeps the level about the same. The limiter is a lookahead limiter with a ceiling you set; behind it sits a hidden soft clipper (2x oversampled) so peaks stay at or under 0.97.
• Presets: 11 factory presets (each carries the three envelopes). Your own presets save the settings and all three envelopes to the Studio 87 presets folder. The Lock button blocks edits to the envelopes.
• Knob and strip changes reach the sound within one audio buffer; Amount, band amounts and Drive are ramped over about 15 ms so they never click.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.